

ActInf Textbook Group ~ Cohort 5 ~ Session 7

(Chapter 3, Part 2) ~ 10/31/2023

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all right greetings everyone welcome back it's Halloween 23 and we're in our second discussion on chapter 3 so we will look at the cohort 5 questions on chapter 3 um so I encourage people to add even now questions directly to cohort 5 questions um or the or or or there's all this other questions so maybe people added directly to the table and so it doesn't have the cohort five tag but before we go to any questions does anyone just want to bring up some piece of chapter 3 that stuck with them or highlight some other topic or share a little bit about how they're seeing the low road and the high road now that we're getting to our second discussion on the high road yes Susan then anyone else who raises their hand so does the low road and the high road are they differentiated by active inference versus Active Learning no Active Learning was the term used by that one paper just to sort of spin the fast ball a little bit different to highlight kind of like a regime that an active inference agent could be in instead of a purely confirmatory sensory mode rather a mode that that reduces uncertainty maximally about world States but active inference is the bigger Circle which what the authors characterized as active learning is a smaller subset Within which incorporates it um and um so Dean um brought up that active inference was a process of provisioning can you speak to that we would have this was in private conversation so we'd have to know a little more about what he meant um but but it's a very evocative yeah so that but that that kind of brought the question of differentiating active learning from active inference versus belief updating so because learning would involve belief updating right yeah as as with a huge number of the topics that we hear about belief updating attention learning anticipation there's often a narrower technical sense like belief updating is just any basian parameter updating at all so belief updating doesn't always refer to a human with a psyche undergoing some kind of like narrative development it can be a lot more narrow and small and less loaded with like philosophical or cultural baggage than if we were to say Hey when's the last time you had a belief updating most people aren't going to say Well when one of my proprioceptors in my arm

you know did something different um so sometimes yeah the narrow sense or the technical sense of a term and then the the broader um um scope of what it might evoke and and so these are all good comments um Kristoff then anyone else who raises their hand uh I I just want to say that the high road and and sort of the kind of um uh you know the road to active inance through this ideas of self-organization and self- evidencing that that really reminded me of um some work that I'm not familiar with uh deeply mathematically but conceptually seems very similar that's um German England's work on the thermodynamic origin of life he kind of um in his work he has shown that if you have um uh system that's receiving an energy from from external source and it has a place to dissipate that energy and there are certain other conditions that need to be met then the system will necessarily organize itself in such a way uh to take advantage of that energy source um so it will start creating structures that will allow it to um um to uh you know utilize that energy and in his talk I think that was at MIT a couple years ago you know at the very end of the talk very kind of he makes this offhand remark that probably um when the energy coming in has some kind of temporal structure that may eventually lead to the emergence of intelligence or something like that that kind of made me when when when he said that in talk made me think about you know faradise speculating about the electromagnetic nature of light uh he's he's probably right and I'm pretty sure that he's I'm quite sure because he started with thermodynamics with his work that it can be recog consed um with active inference so it's also kind of fascinating that there is this very close but slightly different view that kind of arise arrives at pretty much the same um same kind of conclusion from you know um slightly quite different set of assumptions but related yeah very interesting Dr Einstein than anyone else um this isn't low road High Road specific so if you want to shelf it for a second but I was curious about something on page 52 that goes back to the conversation we were having here um two weeks ago in the in the morning group or wherever you guys are at and it's about the idea of perception is action um I remember we were getting into that and on 52 you know they um use this example of if you're hungry and your your glucose is low so you have a hunger response and so you know there's this line that says um perception cannot go further than to reduce your surprise and it got me thinking about like if anybody here is ever fasted for for example you know you'll be really hungry the first day or whatever and then by the second day your hunger is just dissipated it and that that got me thinking about that conversation we were having here regarding perception is action because you know what is it in terms of this field I mean obviously that's like a belief updating about your need to eat or something but what what would be the mechanism you know uh

according to this theory that would cause your hunger to dissipate because there's been no act there's been no action in the external world to reduce your hunger it seems like it would be purely just a matter of perception and so when when they have that one line that perception you know can go no further than to reduce the surprise I yeah it just made me think about that kind of as a thought experiment cool yeah I know a little bit about that this is andw okay go for it go for it yeah there there is action going on your body's metabolism is changing and um there is a yeah there's there's essentially external states that are functioning on your mind um in that transition between day one and day two on fasting that makes it not a completely internal State derived activity I mean would that count as taking an action though I mean because you're not consciously you know changing your external environment I think that if you sketch it according to the active ontology it'll bring it into the rails if you sketch out what's being observed what are the hidden States being inferred it will clarify a lot of pie because otherwise I mean we could go infinitely into the actual biomolecular processes that happen in and fasting but to kind of return that's not my question yeah yeah but that's everything is how how do we take that situation of Interest with its particularities and approach it from The High Road Francis and then and then Kristoff so uh from counseling background I'd say that you have to look at the whole self the part of you that needs the need for food is one of your needs you have other needs maybe the need to uh fit in with your uh Group whatever need is driving you to fast um so you have to look at uh the perceptions goals of the whole self and how these balance out uh you're always dealing with conflict in a conflict we have multiple needs at any one time um so uh uh yeah um and how that would work at the Active inference level I'm not entirely sure maybe it would uh reinforce the uh ex uh strengthen uh the expectations in such a way that uh you ignore the um evidence against uh I don't I don't have a sufficiently good strong understanding to uh propose the mechanism that's what I wanted to say it's the whole self cool Mark Miller has explored this kind of like multi-drive balancing and about well-being as optimizing well across multiple domains because you can't just like optimize your blood glucose and not not your blood oxygen you die um and then also internal or covert action as attention to Simply disend or to attenuate from the grumbling of the stomach helps make it an unsurprising path of policy selection not to order something at the restaurant or put the fork in the mouth or whatever it may be um Andrew then Kristoff sure uh not just a general thought of um regarding the active inference framework I mean could we say something like um you know someone is making a decision to fast is a sort of like a higher order decision but nonetheless on the day today if they're not used to fasting or even if they have in the past but

you know have not been for say the past week such that they are very used to and therefore expecting to eat at regular timing intervals on the day today that despite the fact that they've made this kind of higher order decision to not eat uh nonetheless like they their body uh and other processes are expecting food um they are in fact not receiving food because of the fasting and so there's a higher level of something like there's an increase of something like surprisal or or uncertainty that is like even though I can kind of consciously think like oh no I could eat food I'm just choosing not to my body is still experiencing physiological shifts and changes because I'm not eating that is generating a lot of like um kind of uncertainty in my body you know this is different this is not usual this is not realizing my typical set of preferences um and then over time as the body kind of adjusts to that um suddenly the need you know hunger other um kind of sensations of surprise or otherwise might start to reduce is you the body starts to realize like oh no I'm fine like my my homeostasis is not exactly in danger here even though you know it was kind of a shock initially to start experiencing the um you know the the the difficulties of of fasting uh in the first place on day one right so yeah here we have the glucose observations and then you can infer what the hidden state is again it's the modeler's choice map not territory but this could be risk of death so it might be a matrix learning oh actually I don't have a high risk of dying even when it's at this value or not then you mentioned like habits I mean oh it's 5:00 pm so the generative models in the business of the path of least action doing the likeliest thing so if 99% of the last thousand to 5 PMS you did something your prior's high on simply doing it as a matter of habit or of having a preference for high glucose would then lead to the selection of policies that make those observations unsurprising so once you start to kind of take a figure or some formalism and go beyond just associating components of the territory to bringing them in to stitching together a total generative model you find that there's there's many generative models you can draw sometimes with convergent insights other times that highlight different features and that's part of the iterated process of modeling is like developing a portfolio of generative models like many maps of the city and then finding out well where are these special cases or generalizations of each other what's the kind of measurement or experiment that as a modeler is going to help me differentiate amongst these two different structural hypotheses Kristoff then anyone else um you kind of meet me to the punch because I was about to say that it's it's all about tuning well it's not all about but it's largely about tuning Precision parameters um so if you if you do an experiment where you fast regularly like say every week you're going to find it that you know even though you're you're starting from effectively the same starting

point so it's it's a somewhat reproducible experiment week after week after week after week it's becoming easier and easier and easier to fast and also you know like I'm not I'm not an adept meditator but I have some level of skill and in my case um I can easily distinguish between two kinds of hunger there's a intellectual craving for novelty which I I derive quite a lot of novelty from food it's a it's a pretty easy source of novelty and a lot of people I think um uh you know like they feel this uh they're feeling of hunger is kind of virtual that way um and there is a physiological hunger that's that's a very distinct and different kind of feeling um and most of the time I actually know that my body does not need food even though I I feel like I would like to eat something because I'm creating this um this intellectual novelty and I think there's an interesting way to actually experience this very directly like how this process of tuning Precision parameters um happens is that if you take you know a big exhale and empty your lungs and hold your hold your breath for as long as as you can at some point your body will for for you to take a breath you can't really just um you know suffocate that way because a mdala will kick in and they will trigger you know this kind of panic fight ORF flight response that will kick you into taking a big breath but if you do it again and again and again and again very quickly you just get used to it and you're no longer reacting to to that with panic and you can uh hold your breath for much longer and so on so forth and you can actually experience some really nice effects of um saturating your blood with CO₂ which which is which is extremely relaxing um and it pushes like O₂ deeper into the the cells and actually helps to pry oen um away from you know um red blood cells and so on so it's and it's it's entirely controlled by amygdala and that's why it comes with this like a fascinating reaction and you can just tune your you can observe yourself tuning that that parameter and very quickly you can get used to much higher concentrations of CO₂ in your in your blood and it's not a problem at all okay cool just to sort of look ahead to the kinds of generative models that makeup doing active inference modeling we see this all the time figure 4.3 we're going to revisit chapter 7 so this is the partially observable Markov decision process now as shown here there are no Precision parameters but then quickly after they show well yeah you can actually have another parameter kind of on top of each of them little D on top of the D prior little a on top of the a little B on top of the B and so on and this speaks to the extensibility or the modularity of building generative models using base graphs is in the kind of plain first level model you just have a fixed a Matrix and then you can say well what about having a variance on the a matrix like an like a kind of precision on a well if that helps you get at the phenomena that you're interested in that might be something to try you could also continue to even

develop further say well what if we had policies on which attention we wanted to have on a no one's stopping you it's just map building and especially if it's um being done as a kind of thought experiment outside of reconnecting with the empirical measurements then yeah there's basically no end to speculating about where you could it's just like building Lego constructions because you could you could have another variable connected to here and here and here and here and so that's hence it can be kind of a open-ended and aesthetic journey to make these kinds of glyphs but then if you want to reconnect it back to the empirical data which is discussed most in chapter nine then you kind of have a Divergent convergence iterative process where you expand the portfolio of generative models you have and then you use some experiments or measurements to then select or do Bayesian model selection to go back down to a smaller subset of them and then you can kind of diversify from there select them back down um any other thoughts or we can look at some of the most um uploaded questions or any other questions that people want to look to any any other thoughts or or just remarks on the low road and the high road um I I did uh just kind of want to throw it out there like near the end of the high road like box 3.3 they're describing entropy minimization and open-ended behavior and I guess just I just want to mention generally the topic of like how does an agent within a fluctuating environment uh on the one hand uh find means of like adapting to that environment like that is a h how does it act in order to maintain something like homeostasis or allostasis but at the same time not become too rigid like not repeatedly make the same exact decisions purely to maintain its its homeostasis and instead uh allow for something like improvisation or um like they phrased it as an open-ended repertoire of Novel behaviors creativity or something like that can arise so just that question of like Behavior being like too rigid in an an environment versus like actually being able to change being able to come up with new things unexpected things even things that haven't been tested so you would expect that like why would you do that that would probably bring higher surprise uh even more uncertainty so that just those kinds of questions balancing those things yeah the the there's there's a ton to to say there [Music] um well what one aspect is in expected free energy so in the Criterion used for prospective selection of action there's an Information Gain or epistemic value term so surprising information informative information is actively sought after because policies that lead to Information Gain are directly chosen so that's one kind of prospective looking answer um that holds up both seeking observations that that Accord with um preferences that's loaded on to pragmatic value and seeking out informative information that's the information game and I and that's um referenced here at the end of the Box um another

piece to it is it's kind of like all these different seesaws epistemic and pragmatic value or underfitting and overfitting active inference isn't saying there is a specific answer to that but it gives us the the canvas that we could then do the balancing within like you can make an agent that only undertakes epistemic action or conversely under only undertakes pragmatic action so it's not that just framing it within active inference gives you an Adaptive agent it's just that this is an expressive enough framework for generative modeling that you can find those special regions or manifolds within State space that might have those kind of like flexible or adaptive properties so it's not that something works and and does balance informational and pragmatic value just because it's using expected free energy it has to be kind of tuned to to to sit at that special place Dr Einstein then anyone else okay you're unmuted I never expected these computers you know still learning curve since I've hopped in the time machine um but um yeah I think last week we we towards the end of the session um got to that that line in box 3.3 this could be understood as the imperative to explain things as accurately as possible but also keep options open and avoid to committing to any specific explanation unless this is necessary that's the maximum entropy principle um I mean it just yeah that's such a curveball they kind of sneak in there at the end I'm still trying to reconcile that because it makes sense um intuitively in respect to like information forging behavior um the necessity to to calculate surprise um Beyond just simply minimizing it or you know in in respect to expected free energy rather than variational free energy so it's kind of I guess an episte it's a it's an epistem ially in interesting question as far as what is the conception of of Truth in the active inference framework I mean it it seems like it implies that everything you know any agent has to Simply view the world and truths in the world through a probabilistic framework and um there can be no absolute you know a priori truth in in this or am I I mean these These are huge huge philosophical questions yeah suffice to say that it's not about what the territory is or does however we may take a probabilistic approach to modeling A system that has deterministic or probabilistic characters and then whether or not we think this or that about this or that and whether it thinks this or that any one of those could be right or wrong or literally whatever so I think it's it's a jumping off point for some larger philosophical questions but certainly it's not at the fundamental layer of the framework to to have an opinion on stances that even with little familiarity we can see one could jump one way or the other um Andrew yeah sorry just to follow up on that um and then my question would be is you know how how would you how do you and maybe I just need to research more but how to reconcile you know that foring imperative or maximum entropy imperative with the with the hamiltonian you know gradient

of least action right is there a I mean is there a statistical means of reconciling reconciling those Concepts yeah great great question the path of behaviors that the agent under goes is being conceptualized as a path of least action most likely thing that the generative model could do so depending on the generative models setup the most likely thing for it to do might be engage in exploratory behavior um Andrew then Colin yeah um we had the what is the low road versus The High Road question earlier um and I thought I'd give my stab at it um nice the low road is the fundamental question is what's the nature of inference and the high road the fundamental question is what is the identity that's my stab I have some um the for follow on questions um I voted up a couple that I thought are interesting and were questions I had um hopefully they raised to the tap top but that's all cool that's awesome because inference could be about anything it doesn't have to be about identity so we could do Bas theorem basian statistics doesn't have to be about a cybernetic autopoietic entity and then identity especially if we're going to be talking about Persistence of identity and self-organization in a dissipate of KNE and all of this it doesn't tell us how so I think this is an awesome um diet um Colin then Pablo I was just stimulated to think from looking again at the um the explanation of expected free energy as uh a balance between uh Information Gain and pragmatic value and I found these these these yeah all these formula were were interesting in that they they all represent a kind of balancing of two things one of which you could prioritize more um but only by reducing reducing the other and anyway it it just seemed it seemed to me that yeah that we're prescribing the right way in which these two things are to be balanced against each other uh when active inference is done and we were kind of saying therefore that this is the way it must happen but uh I was wondering why is it not possible to take a step back and say well my model about how I decide what to do is I could give greater priority to Information Gain and less to pragmatic value and uh is it not a reasonable thing that people could make choices to do do it differently um in what way is forbidden or made Difficult by active inference Theory does that make yeah these These are great questions um very nice Insight that that there's like two pieces broken out in each of these framings um don't don't get hit with the equal time fallacy when you actually calculate this for real policies this one might if if if depending on how the numbers are this might equal 0.1 and this might equal a million so policy selection might be driven by one or the other just because there's two terms doesn't mean that it's 5050 so if the C Vector to be more concrete here is very large and we'll see this later in the textbook if the C Vector has a large size then pragmatic value dominates expected free energy if the C value is small relative to uncertainties then Information Gain dominates pragmatic

considerations and so again this is like a seesaw and then adaptivity is like the Seesaw being balanced but just the Seesaw itself is not balanced because that's kind of there's no one on it but when you actually build the generative model and parameterize it then you'll start to play around and see oh maybe something needs to change because it's only doing information foraging it's never going for the pragmatic outcome um I was thinking more in a situation where they were they were closely balanced and that would be that there was a fine tuning of moving more more more waiting one more than the other it seemed like it's been prescribed here how you should wait them in that situation uh and should I spend time getting results or should I abandon getting results uh and go and get more information uh sometimes we actually face that choice do I do more research or do I start writing now and both seem both seem attractive so both have the similar um weight as it were in the decision-making context yeah this is a uh articulation but not a waiting and and there are policies and niches where maybe a given policy has high pragmatic and informational value so just it if it really if you sketched out the generative model of doing research as some have and when to start or stop or move to a different resource then this articulation would help at least kind of reckon with that and then if it turns out that there's two policies that have a similar EXP expected free energy because one of them is really informative and one is pragmatic then they would have similar posterior probabilities of being selected and that agent would potentially switch between those policies um Pablo then sus yes this chapter it make me think about um that we live in in an entropy uh universe but active uh agents are like entropy Fighters that's kind of right yeah anything that's being repeatedly measured at least has to weather the storm even if it's just an inner particle and then things that are active seem to go even beyond that like they reconstitute themselves and they persist even more than you'd expect for just something and being iner and weathering the storm what does that make you think about it's it's fun it's like uh we are like um as far as we know not many of these agents on on on the universe so I wonder why we it's like uh you know entropy all around us and and we we're trying to like uh who we want it or not fight fight against it and it's it's cool okay yeah that's that's that's classical High Road that's kind of the the mystery of persistence or self of just perpetuation and then why active inferences in the middle is we have a first principles persistence imperative with an implementable process Theory coming from the low road and that's why the active inference generative model can be used to address those kinds of situations of persistence and it also has a tractable way otherwise it's just a description of the problem of persistence without a way to generate any kind of agent to actually do it um Susan then Kristoff um so yeah I like the

distinction between um the nature of identity nature of inference um and that kind of brings me to how to um understand and yeah the probability versus possibility kind of comes into my frame of reference because if I'm looking at the high road through the nature of identity then that's going to lead to you my preferences and my um and my perceived importances so how does that change my Markov blanket how does what change it um if I am if if I'm moving between you know what I'm what I'm interpreting here is that if I look at the high road as the nature of identity then that for me has more to do with the area of possibility or you know going back to a statement made um I think last week in terms of action starts with um an imaginative process or predictive representation so that's going to that's going to drive my perceived affordances right yes what what it there's a lot of different ways to say it for something that's simpler or more sophisticated but yeah what actions can I select from what are my affordances such that I'm in the business of finding out what I am or confirming what I am and confirming what I am might be I'm the kind that goes and finds out new information that's the information foraging piece so just self-evidencing doesn't mean just Naval gazing or just um shutting down sensory input as people for years have kind of repeatedly raised self-evidencing can be broader than just seeking confirmation of sensory evidence in a short term right and and so how would that would that not you know let's say that I recognize that you well hey I could be this way too so I basically um invoked or updated my beliefs about what my affordances are would that not change my Markov blanket expand it am I not thinking depend it's a little it's a little understated because a markup blanket in what sense I mean the sensory apparatus is not going to change so let's just say someone says I'm not the kind of person who goes to parties if it if it were at a party it wouldn't be me so that's the kind of identity and what something expects to be but then if someone has a different internal model says I'm the kind of person that sometimes is out of party and sometimes isn't then they could self evidence in either of those settings but in either of those settings it's they would still have visual input and hands so their their Markov blanket wouldn't change or a generative modeler modeling the person going to the party or not it doesn't need to entail a difference but you could make a situation where the markup blanket does change like I'm the kind of person who does or doesn't use neural link or does or doesn't take on this extra sensory technology because that is a more structural modification of the interface of the agent with the niche and that has a more vital relationship with the blanket and with the object's um sense of self perhaps so can I ask one more question does that so does that lead me to the boundary of my Markoff blanket if I'm updating my affordances sketch out the space and see what it looks like and sketch what is

the bound what is the bound and what's what's inside and what's out outside because marov blankets very general any base graph is going to have marov blankets a plenty so then it can be helpful to clarify are we talking about like what kinds of spaces okay um that's the modeling part right yeah yeah okay R go for it hey Susan so your question about um the nature of identity and how can the Markoff blanket of an agent shift around um reminds me of some ideas that I was thinking about recently and this is maybe a bit more of a psychological and uh philosophical uh perspective but thinking about maybe two kinds of of um organisms let's say um biological cells and then also people uh humans that are in a relationship so um the question is about what do we consider at any given moment to be me what is it that I uh say or or have in my mind this is what counts as me and I think you can notice uh if you get into very serious relationship uh romantic relationship or or even having a child uh I imagine and I think I experienced this as well that the that boundary between what I think of as myself um begins to extend and begins to wrap around uh the other person um and to the extent that we have uh begin to have the same shared goals and beliefs that um the extension of the boundary becomes more well well defined and more uh robust and and the other way that I was thinking about this is in in cells in biological cells how um it seems to be that uh for example in cancer cancer cells uh are cells that have separated from the pable organism have become alienated in in a sense and no longer um share the same goal or or know what's the same um have the same instructions um uh as as the rest of the cells in the organism have so they see the rest of their environments as as external environments whereas uh usually yourselves even though each of them are individuals um they kind of are have this Synergy because they're all integrated and have the same uh sort of sort of objective but um when a cancer cell kind of gets uh alienated and it starts to have the same uh intentions and and goals as as uh a cell that's what is uh not part of an organism just kind of an individual cell in the wild would have which is to um find as as much nutrition and as possible and then start to spread and then that's when you have excessive growth of uh of the cell so uh the connection I make there is how um biological cells also have need to be able to figure out this process of redefining what do they think think of uh how do they think of as themselves and they manage to do that um and that's how you end up having a multicellular organism that is composed of many uh individual cells that would otherwise have their own um notion of of of what they have to do in order to persist as as individuals thank you yes self-organization is described in the kind of current here but then there could be composite self-organization with multiple layers so thank you um Kristoff and Andrew quickly and then we'll look at it try to look at a few questions before we end uh I just

want to make two two very quick points one you know um Susan when you were talking about this notion of identity just somehow the um uh Serenity Prayer popped into my head you know go God grant me the serenity to accept the things I cannot change the the courage to change the things they can and the wisdom to know the difference so what exactly constitutes your identity it's a very good question and if you start looking for it um the answer is really not obvious um and you know we we in in my view we draw these um these Mark of blankets around parts of the graph and we say like there was some you know statistical independence of um U certain variables from certain variables and that's that's a useful distinction but it's not actually um it's a kind of in some sense a virtual uh distinction basically Mark of blanket is a perspective um and that perspective can simply be changed from from from one perspective because there's always some kind of hierarchical nested structure and we can adjust it as as needed so the notion of identity is is very fluid and a lot of traditions um you know like like Buddhist traditions for instance um teach you how to look for your identity and you will find that there is no such thing as identity that boundary cannot be really drawn um and you know the the goal of of a lot of these practices to Simply lead it to the realization that we're all interconnected at at some incredibly deep level and if you're if you're trying to draw that boundary you're always going to find some some kind of way in which the boundary doesn't hold but it's still nevertheless kind of useful and to go back to the previous point about this Information Gain versus epistemic value the way I started seeing free energy principles that we start with Bas theorem and which is kind of like this kind of non-computable in in the general case piece of mathematics and then we use it to almost total pretty much not almost but PR topologically carve up um uh that belief updating in all kinds of different ways to give us a different perspective on what what's going on but it's not it's not an implementation prescription it's a it's it's a certain kind of myopic View for the lens of you know probability distributions and base VI and so on so forth kind of geometric View on the world and how it's actually implemented whether there is a um uh whether there is some kind of parameter that weighs um you know um uh information looking for epistemic Value versus um um the other term um that's that that's still you know that's that's kind of hidden by the by the formalism and that's necessarily hidden because the formalism kind of comes up from basically making this distinction up out of nowhere right which just we just take the posterior expand it through base theorem carve it up however we like it group some terms and say like well this is this and this is that and the distinction is in some sense imposed on something where that that that doesn't necessarily uh you know um it's it's not inherent in the object it's it's it's it's an imposed point

of view okay thank you yes Andrew and Dr Einstein very quickly and then let's see if we can go to a question yeah I just have a couple um thoughts on uh some previous points uh one I forget who it was but the the comment about entropy and US kind of be being entropy Fighting Machines I I just had a thought that we've probably all been embedded in this um Milo that treats uh second law of Thermodynamics as like universally in inevitable and I think there's some good evidence due to um like Cosmic expansion and uh some arguments that um contemporary thinkers are making that uh actually order might be more likely um the second law only applies when you have a closed container uh that you'll reach kind of heat death and most of our evidence in terms of the universe is we are in open system all the way around and closed containers are actually the exception not the rule um I'm going to post a couple links to books that sort of talk about that uh what I I first found it from Julian Barbo um who's kind of a a Oddball physicist um he's got a book called The Janice point that he talks about the relevance for that in terms of time and um also I believe Alicia Giro has been is her name pops up in um the Koda our textbook group um the context changes everything um and uh she's got some interesting arguments that kind of touch on that and also touch on the whole Markoff blanket uh dialogue we're having here um I believe you can get her book for free it's unlocked I'm currently looking at it and highly recommend it so far um and finally the whole um related to this entropy question um there's some uh debate going around on a new Theory called assembly Theory um that some people think is just utter boulder dash but uh it's got a new interesting take on uh the the nature of order um and worth taking a look at um so yeah and oh one final thing markof blankets um I I believe and your Markoff blanket changes if uh you've ever used a hammer before and um you just hammer in a nail um you're you're the change between um where the blanket ends and begins um and your body there's evidence that your mind uh incorporates a tool um from a car to a hammer to anything in between and spectacles will change your uh Markoff blank as well thank you yes indeed that's why there's so much discussion of the extended embedded cognitive processes because we can both hold up what the thing is and understand how tools and so on change it cognitively Dr Einstein quickly and then we'll look at one question yeah just real quickly um I posted a link in the chat to a conversation uh in his hippo head on converging dialogues and just on that that notion of embodiment she does a really thorough overview of some of the philosophical underpinnings of the high road as it uh relates to embodiment in the work of Merlo panti and haiger Andy Clark so I really really recommend it um even for non-philosophers um it's it's very accessible so and she really touches on that the the need of or yeah the relationship of the body and en

and the environment you know as an agent for gathering information and model updating respect to Andy Clark's work so check that out guys cool in our last minutes um first if you feel like you have more to say or you have more questions than answers more answers than questions I so heavily and heartedly encourage you to modify the Koda please please add the links there that's how we move together as a niche if you add questions if you add links if you improve links you will learn you will help others learn it's really super important and special um there's a ton of questions of course that we didn't get to but but just even highlighting some of the questions in red these are like epic things that people have asked and and again there's a lot of improvable discourse in a lot of these topics so like I know it's a lot to to um look at and discuss and so we're like all just kind of forging through this massive Niche but no one's going to read all of this so we're always just kind of like finding and leaving things um forward from each other um that's part of the fun let's just look ahead to what's going to happen this was our second discussion on chapter 3 we're going to be heading next into chapter four chapter four is where we get to the generative models of active inference chapter one just background two and three were like the roads into town chapter 4 is the town this is active inference chapter five is going to discuss what's happening in that town in the mammalian Neuroscience District just focusing on a few human or mammal type Neuroscience cases where generative models have been deployed and that'll take us basically to the end of the year um people can add their different ideas and and so on but that's kind of what the first half of the book the epistemic half of the book ends up looking like and then the second half is bringing closer to the modeling and the Hands-On and so on with the recipe starting in chapter six but um yeah it's quite a journey and the low road and the high road definitely invoke very different regimes of attention chapter 4 um is going to be longer and with some more analytical details than the previous chapters so I encourage people to read it lightly um which may be slowly but there are some formalisms and so on that are not vital for understanding in fact they even suggest like you can like skip over this chapter which is very funny that they say that but um we will return next week I'll be really looking forward to the the notes and the questions that people have so thank you everybody see you next time bye thanks a lot guys Halloween everyone